

# Sound Class 01 Flashcards

Generated by TrickFunda

## 1. What is the fundamental physical nature of sound according to the notes?

'%oK'ùM“, ”r ’%oA“%o>“ ””Û5’%o? ‘Y@ ’éB“” -”É\$“ù ’©M“ ”9\$“ò ”Û/“â 9”f ð

### Meaning:

Sound is a form of Energy.

'yM“Y(“ò “ M’É> ‘Y> •ù “ B’¢ 9”%od

### Trick / KD Hack

Logic Link: Energy is the ability to do work. When you speak, your vocal cords do mechanical work, converting kinetic energy into sound energy.

Logic Link: •©0”Û “â “é0”Û/ ‘Y0’%oG ‘Y@ ‘YM“y.’l> “™H–B ’Â ’¢ ,”12’lG “™H•”Â \$”2 ’© ”r 5”1 “” ”™0”Û! ’ù>•)\$”Û

## 2. In what specific form is sound energy carried from one place to another?

'yM“Y(“ò “ M’É> ‘YK •ù “%oM’Y>’, 8”r &”)8“ G “%oM’Y>’, \$’R “ù8 “Y?“i?“yM’ò 0”)\* ’éG•” 2”r “é/“â “é\$“â 9”%od

### Meaning:

It is carried in the form of waves.

•y8”r \$“ ‘yK•” ”r 0”)\* ’éG•” 2”r “é/“â “é\$“â 9”%od

### Trick / KD Hack

Logic Link: A wave is a moving disturbance. Since sound energy needs to travel without permanently moving the actual air molecules to the destination, it uses a wave mechanism.

Logic Link: ’l0•) •ù ‘y\$“ù.“é( “Y?“YM“yK’Ò 9”%od ‘©B• “ò ””Û5’%o? •©0”Û “â ”2 5“é8”Û\$“Y?“R 5“é/” ’9A’9 ‘YK ‘y’

## 3. What does the term 'Vibration' exactly mean in the context of sound mechanics?

'yM“Y(“ò /“é ’lM“ ?‘Y@ ‘YG “%o ’i0”Û- ’éG•” y •)\*,r ...f–’ ation) “i,”Û& ‘Y> ’ @‘RÛ ” ‘YM’ù> •Y0”Û% “™H

### Meaning:

Vibration means Oscillation.

‘Y ’©( ‘Y> •Y0”Û% “™H ’iK“)( (Oscillation)–@

### Trick / KD Hack

Logic Link: Oscillation is the repetitive variation around a central point. A vibrating guitar string oscillates rapidly, transferring this rhythmic energy to the air.

Logic Link: ’iK“)( •ù ‘YG•)&”Û0” / ’É?•)&” ”r “é0”1 ’90 ’iK“™0“é5’i>“ –“ù(”Û(I> “™H–B “ù “é0 ‘Y> ‘Y ’©( ‘Y0’%oG

## 4. What is the FIRST necessary condition for the production of sound?

'yM"Y("ò 'IM'©("Ù( 'Y0'%G 'YG ")?•ò \*TM2" "Y6"Ù/'R 6" M'B "Ù/"â 9"f ð

### Meaning:

Vibration of an object.

'Y?"%@ "Y8"Ù\$" "â •)\*%d

### Trick / KD Hack

Logic Link: Energy cannot appear from nowhere. A physical object must vibrate (oscillate) to act as the source of kinetic energy that becomes sound.

Logic Link: •©0"Ù "â "TM@•" 8"r -" \*"Ù0'Y '%9" "TMK "% 'I@-B "ù8" -"É\$"ù "Y8"Ù\$" "2 'I?'Â " M'É> 'YG "%M

## 5. What is the SECOND necessary condition for sound to travel?

'yM"Y("ò " /"é\$"Ù0"â "r 2"ù 'iB"%0" "Y6"Ù/'R 6" M'B "Ù/"â 9"f ð

### Meaning:

Presence of a Medium.

'é>'yM'ù. 'Y@ •TM\*"%"M'Y?'I?=@

### Trick / KD Hack

Logic Link: Sound energy is passed along like a relay race. Without particles (a medium) to bump into one another and pass the baton, the energy stops immediately.

Logic Link: 'yM"Y("ò " M'É> " ?")G 'iL' < 'Y@ 'I0"" 'yG 'É""É\$" 9"%d •ù -'iB"%0"r 8"r 'Y0"é("r " ,"% ' , \*"é8 'Y0'%"

## 6. What are the three possible states of a medium through which sound can travel?

'é>'yM'ù. 'Y@ "YG 'I@', 8•)-"é5"ù\$ •Y5"%M'Y>•ù 'YL', 8" 9"% "É?"%8"r 9"1 " ""Ù5'%? 'ù>'IM" > 'Y0 "%

### Meaning:

Solid, Liquid, and Gas.

' K", ...6öÆ-B'Â &"Ù0"R „Æ— V-B'Â " "%8 (Gas)—@

### Trick / KD Hack

Logic Link: Since sound requires particle collisions, it will travel through any state of matter that contains interacting particles.

Logic Link: '©B• "ò ""Ù5'%? 'YG ")?•ò '9K•" " 'YM'Y0 •i5"iM'ù "TMH, 'ù9 '©&"é0"Ù% 'Y@ 'Y?"%@ 'Ù@ •TM8 •Y

## 7. If an explosion occurs on the Moon, does the sound reach Earth?

'ù&"ò •)&"Ù0"é> '©0 'YK•, 5"ù8"Ù+"1 "TMK'I> "TMH, 'IK 'YM'ù> 'yM"Y("ò \*%9%"Ù5" \$'R \*TM A• 'I@ "TMH?

### Meaning:

No, it never reaches Earth.

'%9" , 'ù9 '©C'YM"Y@ 'I 'Y-" ("TM@•" \*TM A• 'I@ "TMH=@

### Trick / KD Hack

Logic Link: The vast distance between Earth and the Moon is filled with space vacuum. Sound requires a mechanical medium, which ends at the Earth's atmosphere.

Logic Link: '©C'YM"Y@ 'I0 '© 'iM" .“â ”r ,” 'Y@ “Y?”i>” &”)0” •)\$“ ?'YM“r (“ù0”Ù5“é\$ “%G 'Ù0” 9”%d 'yM“Y(“ò

## 8. Why exactly does the sound of an explosion on the Moon fail to reach Earth?

‘© 'iM“ .“â \*” 9” “Y?”%M’1K‘ò ” ””Ù5’%? '©C'YM"Y@ 'I 'YM'ùK•" (“™@•" \*”™A• '©>'I@?

### Meaning:

Because there is a vacuum between the Earth and the Moon (no medium).

‘YM'ùK•) “ò \*”9%”Ù5” “ •)&”Ù0‘é> ‘YG ‘É@‘¢ (“ù0”Ù5“é\$ (vacuum) “™H (‘YK•, .“é”Ù/’â (“™@•”™d

### Trick / KD Hack

Logic Link: A vacuum means zero particles. No particles = no collisions = no wave propagation.

Logic Link: '%?’“ M“Y>’B “â “ M'R 9”, 6”)(”Ù/ ‘Y#–B ”1 ‘Y# '%9” = ‘YK•, ‘YM'Y0 '%9” = ‘YK•, \$“ ‘r \*”Ù0“%>” (

## 9. If an explosion happens on the Moon, will you be able to see the flash (light) from Earth?

‘ù&“ò •)&”Ù0‘é> '©0 ‘YK•, 5“ù8”Ù+”1 “™K'I> “™H, 'IK ‘YM'ù> •i\* '©C'YM"Y@ “%G •™8'Y@ ‘©.'R %\*”Ù

### Meaning:

Yes, the light will reach Earth.

“™>• Â \*”Ù0‘Y>“b \*”9%”Ù5” \$’R \*”™A• ”y “éd

### Trick / KD Hack

Logic Link: Light operates on a completely different physics principle than sound; it uses self-sustaining electromagnetic fields.

Logic Link: '©M“ “é6 'yM“Y(“ò ” \$” 2’%> ‘éG•” \*)0” \$“ 9 “%G •Y2'r -”É\$“ù ” 8“ù&”Ù“é 'B \*” “é. ‘Y0'I> “™H; ‘ù9

## 10. Why does light from a Moon explosion successfully reach Earth through a vacuum?

‘© 'iM“ .“â ”r 5“ù8”Ù+”1 ‘Y> '©M“ “é6 '%?’“ M“Y>’B ”r .“é”Ù/’â 8”r 8’12'I>'©B“ M“Y '©C'YM"Y@ 'I 'YM'

### Meaning:

Because Light does not require any medium to travel.

‘YM'ùK•) “ò \*”Ù0‘Y>“b ”2 /“é\$”Ù0“â “ (“r ”r 2“ù ‘Y?”%0@ ‘é>'yM'ù. ‘Y@ •i5“iM'ù 'I> '%9” “™K'I@ “™H-

### Trick / KD Hack

Logic Link: Light is a Non-Mechanical wave. It generates its own alternating electric and magnetic fields to move forward, completely independent of physical atoms.

Logic Link: '©M“ “é6 •ù ‘yH“ Ù/“é 'iM“ ?’R \$“ ‘r 9”%d ‘ù9 •i ”r ,’)<%G ‘YG “)?•ò '©(“â 8”Ù5’ù ‘Y> '©M“ \$”Ù/“é5”

## 11. When observing a distant Rocket launch, which do you observe first: the shining flash or the roaring sound?

'iB" 8"r 0"™ "y ")I'‰M'¢ &"y 'IG "‰. 'òÂ '¢ \*™2"r "Ù/"â &"y 'IG "™H•# '©.'Y\$" 9" " K"i(" /"â " M'É("â

### Meaning:

You see the Shining (flash) first.

•i\* '©9")G '©.'R ...6†–æ–æ'r &"y 'IG "™H•)d

### Trick / KD Hack

Logic Link: The visual and auditory events happen simultaneously at the source. But they travel to the observer at vastly different velocities in the air.

Logic Link: 'iC"iM'ò " 6"Ù0"YM'ò 'ù("é • 8"Ù0"1\$ '©0 •ù "‰>'R 9"1\$" 9"‰ –B 2"y "ù( "YG "™5"â ."y 'Y>'1@ •Y2

## 12. When observing a distant Rocket launch, which do you observe second?

'iB" 8"r 0"™ "y ")I'‰M'¢ &"y 'IG "‰. 'òÂ '¢ &")8" @ 'É>" ‰,"é& 'éG•"" "Ù/"â &"y 'IG/"‰A'‰\$ "r 9"‰ ?

### Meaning:

The Sound of the rocket.

" I'YG'ò " "'Ù5'‰? (Sound)–@

### Trick / KD Hack

Logic Link: Sound relies on physical air molecules bumping into each other, which is a slow mechanical process compared to photons flying through the air.

Logic Link: 'yM"Y("ò –"É\$"ù "Y>'ùA •Y#" •" "r 'RÙ&")8" G "‰G 'ù " >'‰G '©0 '‰? " M'Ù0 'Y0'I@ "™H, 'ÉK "™5"

## 13. What is the physics reason you see a rocket shine before you hear its sound?

•y8'Y> 'ÙL'I?'Y@ 'Y>" # 'YM'ù> "™H 'Y? •i\* " I'YG'ò " "'Ù5'‰? "‰A'‰("r 8"r †™yÇ •™8'Y@ '©.'R &"y

### Meaning:

Because the Speed of Light is much greater (>>>) than the Speed of Sound.

'YM'ùK•) "ò \*"'Ù0'Y>"b " 'I? 'yM"Y("ò " 'I? "‰G 'É9" \$ •Y"ù (>>>) "™K'I@ "™H–@

### Trick / KD Hack

Logic Link: Light is the fastest phenomenon in the universe. Sound is relatively slow. Light wins the race to your sensory organs by a huge margin.

Logic Link: '©M" "é6 'ÉM" 9"Ù."é ' " 8'É8"r \$"y "Â 'ù("â 9"‰d 'yM"Y("ò '©G'YM"y>'YC'B "'." 9"‰d '©M" "é6 •i

## 14. What is the standard metric used to measure the Loudness of sound?

'yM"Y("ò " \*"'Ù0'É2'I> (Loudness) 'é>'©("r "r 2"ù 'Y?,".é('R ." "Ù0"ù 'Y> •™\*'"ùK'r "ù/"â "é\$"â 9"fđ

### Meaning:

Decibel (dB).

' G"%?'É2 (Decibel)—@

### Trick / KD Hack

Logic Link: Loudness relates to the amplitude of the wave. The human ear perceives amplitude logarithmically, so scientists use the Decibel scale to map physical intensity to perceived loudness.

Logic Link: '©M" ," )\$ "â \$ " 'r "r 'ù>'â 8"r 8•),•) "ù\$ "™H—B ."é("R "é(•i/"é. 'YK ") " '9 " / " B'¢ 8"r ."é('l> "™H, •y8")

## 15. What unit is used to measure the intensity of environmental 'Noise'?

'©0"Ù/"é5" #" / "'iK" r „æö—6R' " "\$" 5"Ù0'l> 'é>'©("r "r 2"ù 'Y?", 'Y>•, "â '©/"1 'Y?'ù> 'É>'l> "™H?

### Meaning:

Decibel (dB).

' G"%?'É2 (Decibel)—@

### Trick / KD Hack

Logic Link: Noise is simply an unwanted or unpleasant sound. Since it is still sound energy, it is measured using the exact same Decibel scale.

Logic Link: "iK" "y5"" 'R "Y>•) "ù\$ 'ù> •Y\*"Ù0"ù/ 'yM"Y("ò 9"%d '©B• "ò /" 'Ù@ 'Ù@ 'yM"Y("ò " M'É> "™H, •y8")

## 16. Mathematically, what is the exact relationship between a 'Decibel' and a 'bel'?

'y#"ù\$" / " B'¢ 8"rÂ y!"y8"ù, ""r „FV6—&VÂ' " y,"y2' (bel) 'YG 'É@'¢ 8"ù@'R 8•),•)' 'YM'ù> "™H?

### Meaning:

A Decibel is 1/10th part of a bel.

' G"%?'É2 •ù 'ÉG"" "â ó 5"é "™?"%oM"%o> "™H—@

### Trick / KD Hack

Logic Link: In the metric system, the prefix 'deci' specifically denotes a factor of one-tenth (1/10). A decibel is one-tenth of the base unit 'bel'.

Logic Link: 'é@'ùM" ?'R "\*"Ù0'9>")@ 'éG•"Â '©8" M'r vFV6'r 5"ù6"y7 " B'¢ 8"r 'RÙ&"%5"y (1/10) 'YG 'Y>" 'YK "

## 17. The unit 'bel' (used in Decibel) was named in honor of which famous scientist?

•y "é "ÉG""r %o!"y8"ù, ""r ."y '©M" /" "Ù\$) 'Y> '%o>'â "ù8 '©M" 8"ù&"Ù" "YH'ÉM'é>'%o?'R "r 8'ém'é>,".y

### Meaning:

Graham Bell (Alexander Graham Bell).

'yM" >"™. 'ÉG"" %o ")G'YM'ÉG•)!" "Ù0"é9"â ,"y2)—@

### Trick / KD Hack

Logic Link: Bell did pioneering work in telecommunications and acoustics. The unit measuring sound intensity was named after his surname 'Bell' (shortened to 'bel').

Logic Link: 'ÉG"" ("r &")0"%o '©>" " ""Ù5"%o?'Y@ 'éG•" 'yM" #" "é. 'Y?'ù>—B ""Ù5"%o? 'l@"YM" \$"â ."é\*"%oG "Y>"

## 18. What is the lower frequency limit of the Audible Sound Range for Humans?

'é(" 7"Û/"¹ 'YG ")?•ò 6"Û0"YM'ò ""Û5'%o? "%o@'é> (Audible Sound Range) 'Y@ '%o?'©2" "YC'IM'I? "%o@

**Meaning:**

20 Hz.

20 "™0"Û "Û (Hz)–@

### Trick / KD Hack

Logic Link: If an object vibrates fewer than 20 times per second, the wave energy is too slow and weak to move the tiny bones in our middle ear.

Logic Link: 'ù&"ò "¹ "Y8"Û\$ "™Û0'I? "%oG'Y ' # 8"r 'â , "é0 'Y '©( 'Y0'I@ "™H, 'IK 'IO•) •©0"Û "â 'I(" "" ." " 'é "

## 19. What is the upper frequency limit of the Audible Sound Range for Humans?

'é(" 7"Û/"¹ 'YG ")?•ò 6"Û0"YM'ò ""Û5'%o? "%o@'é> 'Y@ •©\*" @ •i5"9\$"Û\$ "ò 8" . "â "Û/"â 9" f ð

**Meaning:**

20 kHz (or 20,000 Hz).

20 'Y?(")K"™0"Û "Û "Â †' ‡¢' %o/"â # Ã ‡¢™d

### Trick / KD Hack

Logic Link: If an object vibrates more than 20,000 times per second, the eardrum physiologically cannot react fast enough to decode the signal.

Logic Link: 'ù&"ò "¹ "Y8"Û\$ "™Û0'I? "%oG'Y ' # Ã 8"r 'y?'R , "é0 'Y '©( 'Y0'I@ "™H, 'IK 'Y>', "â \* "M'i> "i>" @ "

## 20. What is the complete Audible Sound Range for normal human hearing?

"%o>'é>%oM'ò . "é("R 6"Û0"Y# 'YG ")?•ò \*)0"Û# "iM" 5"Û/ 'yM"Y("ò 8" . "â „ udible Sound Range) 'YM'ù>

**Meaning:**

20 Hz to 20 kHz.

20 "™0"Û "Û "%oG 20 'Y?(")K"™0"Û "Û "Éd

### Trick / KD Hack

Logic Link: This is the biological evolutionary sweet spot. Our ears evolved to hear frequencies within this band because it encompasses speech and environmental dangers.

Logic Link: 'ù9 'ÉH"Y?'R 5"ù "é8"Y>'i@ "%oM"Y@ "ò 8"Û\*"™ "™H–B 9'é> "G 'Y>', ", , "%o ' "r -" \$" "YC'IM'I?'ùK•

## 21. What is the 'Persistence of Hearing' expressed as a fraction of a second?

•ù "%oG'Y ' "r -"ù("Û( (fraction) 'YG " B'¢ .y "YM'ù "Û\$ "iM" 5'2 " &"9""É\$ "âr ... ersistence of Hearing

**Meaning:**

1/10 seconds.

1/10 "‰G'Y ' d

### Trick / KD Hack

Logic Link: The brain needs processing time. A sound signal lingers on the auditory nerve for 1/10th of a second before fading out.

Logic Link: 'é8"Ù\$ "ù7"Ù 'YK '©M" 8•)8"Ù " # "‰. 'ò " "Y6"Ù/'Y\$ "â 9"1\$" 9"‰d •ù 'yM"Y("ò 8•) "y\$ "iM" 5'2 \$•)\$"Ù

## 22. What is the 'Persistence of Hearing' time expressed as a decimal?

'i6'é2"R †FV6–Ö Â' "r 0")\* 'éG•" 5"Ù/'YM'B y6"Ù0"Y# 'Y@ 'iC')<'l>' 'Y> "‰. 'ò "Ù/"â 9" f ð

**Meaning:**

0.1 sec.

0.1 "‰G'Y ' d

### Trick / KD Hack

Logic Link: This 0.1s threshold is crucial for room acoustics. To hear a distinct echo, a reflected sound must take longer than 0.1 seconds to bounce back.

Logic Link: 'ù9 0.1s "‰@'é> 'Y." G 'Y@ 'yM"Y("ù " "r 2"ù 'é9'IM"Y\*)"0"Ù# "™H–B 'R ") 'yB•) (echo) "‰A'‰("r

## 23. Which specific human organ is responsible for the 'Persistence of Vision' effect?

'YL', 8"â 5"ù6"ù7"Ù 'é>'‰5 •Y 'r y&"97"Ù "ò " &"9""É\$ "âr ... ersistence of Vision) '©M" -"é5 'YG ")?•ò "

**Meaning:**

Retina (in the Eye).

" G'ù?'‰> (•i 'b ."y )–@

### Trick / KD Hack

Logic Link: Light creates a chemical reaction on the Retina. This reaction takes a fraction of a second to reset after the light source is removed.

Logic Link: '©M" "é6 " G'ù?'‰> '©0 " >"‰>'ù("ù '©M" \$ "ù "Ù0"ù/"â , '‰>'l> "™H–B "\*"Ù0'Y>"b 8"Ù0"1\$ 'YK "™ "â

## 24. What is the 'Persistence of Vision' time expressed as a fraction of a second?

•ù "‰G'Y ' "r -"ù("Ù( 'YG " B'¢ ."y "YM'ù "Ù\$ "iC"yM'ù? 'Y@ 'iC')<'l>' 'Y> "‰. 'ò "Ù/"â 9" f ð

**Meaning:**

1/16 seconds.

1/16 "‰G'Y ' d

### Trick / KD Hack

Logic Link: The visual processing system clears frames slightly faster than the auditory system clears sounds, allowing us to track fast movement.

Logic Link: 'iC"iM'ò "\*"Ù0"‰ "‰M'Y0'2 "\*"Ù0'9>")@ "iM" 5'2 "\*"Ù0'9>")@ 'iM"Y>" > 'yM"Y("ù/"1 'YK "‰>'1< 'Y0'‰G

## 25. What is the 'Persistence of Vision' time expressed as a precise decimal?

•ù "% " 'i6'è2"R "r 0")\* 'éG•" 5"Û/'YM'B y&"97"Û "ò " &"9""É\$"âr "â 8'é/ 'YM'ù> "™H?

**Meaning:**

0.062 sec.

0.062 "%G'Y ' d

### Trick / KD Hack

Logic Link: Math conversion:  $1/16 = 0.0625$ , often rounded to 0.062 in physics notes.

Logic Link: 'y#"ù\$ " B'©>•)\$" #:  $1/16 = 0.0625$ , 'É?"%G •Y "Û8" -"É\$"ù " ("1 "Û8 'éG•" ã c" \$'R "12 'Y?'ù> 'É>'I> "

## 26. What is the scientific branch focusing on the Study of the Production of Sound Waves called?

'yM"Y("ò \$" 'yK•" "r 'IM'©>'i( 'YG •Y""Û/'ù( '©0 'yM'ù>', "y 'iM" '?B " ("r 5"é2" 5"% "Û "é("ù "i>'i> 'YK "

**Meaning:**

Acoustics.

'yM"Y("ù " „ 6÷W7F–72™d

### Trick / KD Hack

Logic Link: Derived from the Greek word 'akoustos' meaning 'heard'. It is the comprehensive study of mechanical waves (sound) in gases, liquids, and solids.

Logic Link: 'yM" @'R 6'ÉM'b y 'YL"%M'ùK",r 8"r 2"ù/"â 'ù> "™H 'É?"% "â " M'R 9", y8" ("â 'ù>–B /" " "%8"1 , 'I0"

## 27. Which laboratory instrument is used for the Study of the Waveform of any signal?

'Y?"%@ 'Û@ "% 'YG'B ‡6–væ Â' "r \$" 'r 0")\* (waveform) 'YG •Y""Û/'ù( 'YG ")?•ò "ù8 '©M" /"1 "i>")> •™

**Meaning:**

Oscilloscope.

' 8"ù2"18"Û "1\* (Oscilloscope)–@

### Trick / KD Hack

Logic Link: An oscilloscope plots fluctuating electrical voltages on a 2D screen, translating invisible sound pressure oscillations into visible graphical waves.

Logic Link: •ù •i8"Û "%@")8"Û '¢ \$B 8"Û "Û0" ( '©0 •™\$"é0–'©""É>"R 5"é2"r 5"ù&"Û/" \$ "YK")M'ùG'Â " 8"é "ù6 "

## 28. Based on the requirement of a medium, what are the two broad categories of Waves?

'é>'yM'ù. 'Y@ •i5"iM'ù 'I> 'YG •i""é0 '©0, 'I0•) "1 'Y@ 'iK "YM'ù>'© "iM" G'9?'ù>•" "Û/"â 9"% ?

**Meaning:**

1. Mechanical Waves

2. Electromagnetic (Non-Mechanical) Waves



1. 'ù>•)\$"Ù0"ù 'l0•) "y (Mechanical)
2. "Y?"iM'ùA'B " ."Ù,"Y@'ò ò "%00-'ù>•)\$"Ù0"ù 'l0•) "y (Electromagnetic / Non-Mechanical)

## Trick / KD Hack

Logic Link: Nature bifurcates wave energy. If it bumps atoms, it's Mechanical. If it propagates its own electric/magnetic fields, it's Electromagnetic.

Logic Link: '©M" "9\$`ò \$" 'r " M'É> 'YK 'iM"Y?'`Ù>`É?'B " \$" 9"%o d `ù&`ò /" \*".`é#". •" 8"r 'Y0`é\$`â 9",Â \$"2 /" /`é

### 29. What is the strict defining rule for a Mechanical Wave?

'ù>•)\$"Ù0"ù '10•) (Mechanical Wave) 'YG ")?•ò 8'iM'B \*"'?Ù>"y?'B ("ù/'â "Ù/'â 9" fđ

**Meaning:**

It ALWAYS Requires a Medium.

•y8"r 9'éG"i> •ù 'é>yM'ù. (Medium) 'Y@ •i5"iM'ù 'l> "™K'l@ "™H-@

## Trick / KD Hack

Logic Link: 'Mechanical' implies physical parts interacting. The parts here are the molecules of a solid, liquid, or gas. No parts, no mechanics.

Logic Link: "ù>•)\$"Ù0"ù ' 'Y> 'I>'IM'©0"Ù/ 'ÙL'I?'R -"é "¹ 'Y@ '©0"%M'©0 'YM" '?'ù> "%G "™H-B /"™>• "r -"é '

### 30. What is the strict defining rule for an Electromagnetic (Non-Mechanical) Wave?

"Y?'iM'ùA'B " .ù, 'Y@'ò % " %0-'ù>•)\$"Ù0"ù ) 'i0•) 'YG ")?•ò 8'iM'B \* " ?'Ù>"y?'B ("ù'â "ù'â 9")fđ

**Meaning:**

It DOES NOT require a medium.

•y8”r “ù8” .“é”Ù/â ” “Y6”Ù/“Y\$“â (“™@•” 9”1\$” 9”%od

## Trick / KD Hack

Logic Link: EM waves are composed of oscillating energy fields that self-sustain. They do not rely on bumping physical matter to transmit energy.

Logic Link: •% 'â \$ " 'yG•" &"12'%6" 2 •©0"Ù "â "Ù7"y\$"Ù0"1 "•%G 'É(" 9"1\$" 9"% 'ÉK "%M"Y/•" "2 ,'%>•ò 0'i\$"

**31. Mechanical waves are further subdivided into what two geometric types?**

'ù>.)\$'Ù0"ù 'lO.)"1 'YK.ï"r "ù( 'iK 'ÉM'ù>'é?'l@'ò \*"'Ù0'Y>"K."."y "Y?'Ù>'É?'B "ù/'â "ù>"™H?

**Meaning:**

1. Longitudinal Waves
2. Transverse Waves

1. •Y(" &"%0"Ù"Ù/ 'lO•) "y (Longitudinal)
2. •Y(" \*"Ù0"%0M'R \$" 'yG•" ...@ransverse)

## Trick / KD Hack

Logic Link: These sub-classifications describe purely how the particles vibrate relative to the direction the energy is traveling.

Logic Link: 'ùG •™\*-“Y0”Ù ” “ # ’©B“ @ ’l0“ 8”r 5“ M’9( ‘Y0’lG “™H•” “ò “ M’É> ‘É?“, &“ù6“â.”y ’ù>’lM“ > ‘Y0 “ 9

### 32. In a Longitudinal Wave, what is the specific angle (theta) between the wave direction and particle vibration?

•ù •Y(" &"%0"Ù""Ù/ '10•) (Longitudinal Wave) 'éG•"Â \$" 'r &"ù6"â " '2 •)\*', "r," "Y?"i?"yM'ò "1# ('Y@

**Meaning:**

Angle = 0°

'YK'2 Ò

#### Trick / KD Hack

Logic Link: The wave travels straight forward. The particles also bounce strictly left and right along that exact same straight line (0 degrees offset).

Logic Link: '10•) "%@'yG •i "r ,')<'l@ "TMH-B '2 -" "%@ "%@'y@ " G'i> (0 ' ?'yM" @ ' +"%G'ò' "r 8"é% "%o "Ù\$

### 33. In a Longitudinal Wave, how is the direction of particle vibration described relative to the wave direction?

•ù •Y(" &"%0"Ù""Ù/ '10•) 'éG•"Â '2 •)\*', " &"ù6"â "â 5" M'9('10•) 'i?"i> 'YG "%o>'©G'YM"r "%o8"r "ù/"â

**Meaning:**

Parallel (Left & Right).

"%o."é("é '10 ('É>•ù '10 'i>•ù )-@

#### Trick / KD Hack

Logic Link: 'Longitudinal' implies 'along the length'. The physical displacement of matter is strictly parallel to the energy vector.

Logic Link: '•Y(" &"%0"Ù""Ù/ 'Y> 'I>'IM'©0"Ù/ "" ) 'É>•, "r 8"é% ' "TMH-B \*'i>" M'R "â -"É\$ "ù "Y?"%M'Y>'©( •©0"Ù

### 34. What is the most fundamental natural example of a Longitudinal wave given in the notes?

'%K'ùM", .y 'i?•ò •ò '%A'iH" M'yM'ò \$" 'r "â 8'É8"r , " ( "ù/"é&" \*"Ù0"é "9\$ "ù •TM&"é9" # 'YM'ù> "TMH?

**Meaning:**

Sound Waves.

'yM"Y("ò \$" 'yG•" ...6÷VæB Waves)-@

#### Trick / KD Hack

Logic Link: When you speak, you push a block of air molecules forward. They bump the next block and bounce back (parallel motion), creating the sound wave.

Logic Link: 'É, •i\* 'ÉK")\$ "r 9"%o , 'IK •i\* "TM5"â "r '9A'9 'YG •ù 'ÉM")I'R "2 'yG 'y "y2'IG "TMH•)d "YG •Y ")G 'ÉM")

### 35. Besides sound, what physical object is used as a classic example of a Longitudinal wave in the notes?

'yM"Y("ò "r ")>"Y>, '%K'ùM", .y •Y(" &"%0"Ù""Ù/ '10•) 'YG 'YM")>"%o?'R 'i>"TM0'2 "r 0")\* 'éG•" "ù8 "Ù

**Meaning:**

A Spring.

•ù “%M’©M“ ?•) (Spring)–@

### Trick / KD Hack

Logic Link: If you stretch a slinky (spring) and push one end forward, a wave of compressed coils travels down the spring parallel to the push.

Logic Link: 'ù&"ò '¢ 'R 8"Ù2"ù 'Y@ ("M'©M" ?•) ) 'YK '1H">'IG "TMH•" " 'R 8"ù0"r "2 'yG 'y "y2'IG "TMH•"Â \$"

## 36. In a Transverse Wave, what is the specific angle between the wave direction and particle vibration?

•ù •Y(" \*\*"Ù0"%M'R \$" 'r ...@ransverse Wave) 'éG•"Â \$" 'r &"ù6"â " '2 •)\*, "r ," "Y?"i?"yM'ò "1# 'YM

**Meaning:**

Angle = 90°

'YK'2 Ò “

### Trick / KD Hack

Logic Link: The wave energy wants to travel horizontally across the room, but the particles only bob directly up and directly down, creating a perfect 90-degree intersection.

Logic Link: 'I0•) •©0"Ù "â 'é0"r ."y 'YM"yH'I?'Â 0")\* "%G 'ù>'IM" > 'Y0"%> '©>"TM\$" 9",Â 2"y "ù( 'Y# 'YG"Y2 "%o(

## 37. In a Transverse Wave, how is the direction of particle vibration described relative to the wave direction?

•ù •Y(" \*\*"Ù0"%M'R \$" 'r ."y , 'Y# 'Y '©( 'Y@ 'i?'i> 'Y> "Y0"Ù#, \$" 'r &"ù6"â "r 8"é\*"y "Ù7 'YH"%G 'Y?'ù

**Meaning:**

Perpendicular (Up & Down).

" ) 'É5'B %o '©0 'I0 '%o@'©G)–@

### Trick / KD Hack

Logic Link: 'Trans' means across. The particle displacement cuts strictly across (perpendicular to) the path of energy propagation.

Logic Link: "ùM" >•)8' 'Y> •Y0"Ù% "TMH •i0-'©>" d 'Y# "Y?"%M'Y>'©( •©0"Ù "â \*\*"Ù0"%>" "r \*'R "r " Ù\*"é0 (" )É

## 38. What is the primary natural example of a mechanical Transverse wave given in the notes?

'%oK'ùM", ."y 'i?•ò •ò /"é 'IM" ?'R '%oA'©M" 8"Ù% 'I0•) 'Y> '©M" >'Y."ù '©M" >'YC'I?'R 'i>"TM0'2 "Ù/"â

**Meaning:**

Water Waves.

'É2 'I0•) "y (Water Waves)–@

### Trick / KD Hack

Logic Link: If a boat sits on a lake and a wave passes, the wave moves forward, but the boat just bobs strictly Up and Down. This proves perpendicular (transverse) motion.

Logic Link: 'ù&"ò "1 '%o>"R " 2 '©0 "TMH 'I0 'YK•, 2"TM0 'yA'É0'I@ "TMH, 'IK ")9" 'yG 'É""É\$" 9",Â 2"y "ù( '%o>"R

### 39. Combining its two defining properties, mathematically define the physical nature of a Light wave.

•y8'YG 'iK '©0"ù-"é7"ù\$ 'yA'9K•" "2 ."ù2"é " Â \*'"Ù0'Y>"b \$" 'r " -"É\$"ù '©M" "9\$"ò "2 '9?'I@'ò 0")\* "%o

#### Meaning:

Light is Transverse AND Non-Mechanical.

'©M" "é6 •Y(" \*'"Ù0"%M'R ...@ransverse) 'I0 'yH" Ù/"é 'IM" ?'R „æöâÔÖV6† æ-6 Â' 9"%od

#### Trick / KD Hack

Logic Link: It's Non-Mechanical because it needs no medium. It is Transverse because its electric and magnetic fields oscillate perpendicular (90 degrees) to the direction of travel.

Logic Link: 'ù9 'yH" Ù/"é 'IM" ?'R 9", "Ù/"1 'Y? •y8"r "ù8" ."é"Ù/"â " "Y6"Ù/'Y\$"â ("™@•" 9"%od 'ù9 •Y(" \*'"Ù0"%M

### 40. Name two examples of Electromagnetic (Non-Mechanical) Waves used in cooking and heat sensing.

'i>'%o> '© "é("r " " M'é@ "%o "YG'i( 'éG•" \*'"Ù0'ùA'YM'B 5"ù&"Ù/" \$ '©A'ém'É " / ('yH" Ù/"é 'IM" ?'R' \$" 'Y

#### Meaning:

1. Microwave
2. Infrared

1. 'é>•y "Ù0"15"y5
2. •y("Ù+"Ù0"é0"y! (Infrared)

#### Trick / KD Hack

Logic Link: These are just invisible colors of light. Because they are on the electromagnetic spectrum, they share light's properties (transverse, no medium needed).

Logic Link: 'ùG 'YG"Y2 '©M" "é6 'YG •Y&"96"Ù/ " 'r 9"%o -B ") 'Y? "YG "Y?'iM'ùA'B " ."Ù,'Y@'ò 8"Ù\*"y "Ù "Ù0'â

### 41. Name three examples of Electromagnetic (Non-Mechanical) Waves used in medicine, tanning, and communication.

'©?'Y?'iM"%o>, 'ùH'%o?•) 'I0 "%o '©"> " .y '©M" /" "Ù\$ "Y?'iM'ùA'B " ."Ù,'Y@'ò %o "%o0-ù>•)\$"Ù0"ù ) 'I0•

#### Meaning:

1. X-rays
2. UV Rays
3. Radio Waves

1. •ù "Ù8-" G (X-rays)
2. 'ùB"Y@ 'Y?" #y (UV Rays)
3. " G' ?'ùK 'I0•) "y (Radio Waves)

#### Trick / KD Hack

Logic Link: All of these propagate energy through the vacuum of space at the speed of light, proving they do not rely on mechanical particle collisions.

Logic Link: 'ùG "%o-" \*'"Ù0'Y>"b " 'I? "%oG •Y 'I0"ù "Ù7 'YG '%o?" M"Y>'B "r ."é"Ù/"â 8"r " M'É> 'Y> '©M" 8"é0 'Y

## 42. When a longitudinal wave travels through a spring, what is the 'squeezed' phase called?

‘É, ‘YK•, ‘%A’iH“ M’yM’ò \$“ ‘r 8”Û\*”Û0“ù ‘r 8”r 9”1 “ ” “ \$” 9”,Â \$”2 y(“ù ”1!“ÉG “™A•òr “ # ‘YK ‘YM’ù> ‘

**Meaning:**

Compression.

“% ‘©@’ <’, „6ö× &W76–öâ™d

### Trick / KD Hack

Logic Link: As the energy pushes forward, it physically squeezes the spring coils tightly together. This localized area of high density is the 'Compression'.

Logic Link: ‘ÉH“%G-‘ÉH“%G •©0”Û “â ‘yG ‘É““É\$” 9”,Â /” 8”Û\*”Û0“ù ‘r ”™ “)M“, ”2 6“é0” 0“ù “ B’¢ 8”r ‘R 8“é%

## 43. When a longitudinal wave travels through a spring, what is the 'pulled apart' phase called?

‘É, ‘YK•, ‘%A’iH“ M’yM’ò \$“ ‘r 8”Û\*”Û0“ù ‘r 8”r 9”1 “ ” “ \$” 9”,Â \$”2 y “) ‘i@•) ”r •òr “ # ‘YK ‘YM’ù> ‘Y9

**Meaning:**

Expansion (or Rarefaction).

“Y?“%M’I>“ /“â 5“ù0“)( (Expansion or Rarefaction)–@

### Trick / KD Hack

Logic Link: After a compression passes, the coils bounce back and spread apart wider than their normal resting state. This low-density stretch is Rarefaction.

Logic Link: •ù “% ‘©@’ <’, ”r ” “ (“r ”r “,é&, ‘YI•y2”Û8 “Y>’©8 •™ “) \$”r 9”% ‘I0 •Y\*’%@ “%>’é>’%M’ò 5“ù6”Û0“

## 44. In a spring wave, how do the direction of wave travel and the rebound vibration relate?

“%M’©M“ ?•) ‘I0•) ‘éG•”Â \$“ ‘r /“é\$”Û0“â ” &“ù6“â “ \*”) “é5 ‘Y ‘©( ‘YH“%G “% ‘É ‘y?’B 9”% ?

**Meaning:**

They are Opposite but on the Same Line (0°).

“YG “Y?’©0” \$ (Opposite) “™H•” 2”y “ù( •ù “™@ “ G’i> (0°) ‘©0 “™H•)d

### Trick / KD Hack

Logic Link: The wave energy moves continuously right. A single coil pushes right, then strictly rebounds left. The motion is 180° opposite to the wave at that moment, but still locked exactly on the 0° longitudinal axis.

Logic Link: ‘I0•) •©0”Û “â 2’y>’I>“ &“é • “ “) \$” 9”%d •ù •ù “” ” ’ 2 ‘i>•ù ‘y ”y2’I> “™H, ‘1?“ ’ <“é “%G ‘É>•ù •

## 45. According to the notes, what specific type of wave is produced when playing a 'Dholak'?

’%K’ùM“, ”r ‘%A“%>“ Â y””12’Rr ,‘É>’%G ‘©0 ‘Y?“ , 5“ù6“ù7”Û ‘©M“ “é0 ‘Y@ ‘I0•) •™\$”Û\*’%M’, 9”1\$”

**Meaning:**

Longitudinal Wave.

•Y(" &"%0"Ù"Ù/ 'I0•) (Longitudinal Wave)—@

### Trick / KD Hack

Logic Link: A Dholak is a percussion drum. Striking the skin pushes a flat wall of air forward, creating a compression wave. Because it's a sound wave in a gas, it must be longitudinal.

Logic Link: ")K(" •ù '©0"Ù "Ù/"6', !"Ù0'â 9"%d 'IM"Y "â \*""\*Ù0"™>" " ("r 8"r 9"Y> 'Y@ •ù "%\*""é 'i@"Y>" 'yG "

## 46. According to physics laws, what is the ONLY classification of wave that CANNOT possibly exist?

'ÙL'I?'Y@ 'YG '%?'ù."1 'YG •Y(" 8"é0, 'I0•) 'Y> "Y9 •ù 'é>'IM" 5" M'y@'Y0'2 "É( "%> "™H 'ÉK "% 'Ù5

### Meaning:

Longitudinal Non-Mechanical (or Longitudinal Electromagnetic) wave.

•Y(" &"%0"Ù"Ù/ 'yH" Ù/"é 'IM" ?'R %/"â '%A'iH" M'yM'ò 5"ù&"Ù/" \$ '©A'ém'É " /) 'I0•) —@

### Trick / KD Hack

Logic Link: An electromagnetic wave intrinsically generates its fields perpendicular (90 degrees) to its path. Therefore, it is impossible for an EM wave to vibrate parallel (0 degrees, longitudinally) to its path.

Logic Link: •ù "Y?'iM'ùA'B " . "Ù,'Y@'ò \$" 'r •) \$" ?'R 0")\* "%G •Y\*%G '©% 'YG ") 'É5'B f" !"ù "Ù0" ' '©("r "Ù7"

## 47. What broad category of waves do Earthquake waves fall into?

'ÙB'Y '¢ \$" 'yG•" „V thquake waves) 'Y?“, 5"Ù/"é\*"R 6"Ù0"y#" . "y •i\$" 9"% ?

### Meaning:

Mechanical Waves.

'ù>•) \$"Ù0"ù 'I0•) "y (Mechanical Waves)—@

### Trick / KD Hack

Logic Link: Earthquakes require the solid rock of the Earth's crust to transfer their massive kinetic energy. Because they require a material medium, they are unequivocally Mechanical.

Logic Link: 'ÙB'Y '©K•" "2 '©(" 5"ù6"é2 'y\$"ù •©0"Ù "â "2 8"Ù%"é("é 'I0"ù\$ 'Y0'%G 'YG ")?•ò \*\*9%"Ù5" " \*©!"

## 48. What specific geometry defines a Surface Earthquake wave?

"% \$"™@ 'ÙB'Y '¢ ...7W&`ace Earthquake) 'I0•) 'YK 'YL', 8" 5"ù6"ù7"Ù 'ÉM'ù>'é?'I? '©0"ù-"é7"ù\$ 'Y0

### Meaning:

Longitudinal Wave (0 degrees).

•Y(" &"%0"Ù"Ù/ 'I0•) (0 ' ?'yM" @)—@

### Trick / KD Hack

Logic Link: Surface primary waves violently push and pull the ground forward and backward along the exact direction the earthquake is traveling, acting like a giant spring (0 degrees).

Logic Link: "% \$"™@ '©M" >'Y."ù 'I0•) "y 'ÙB'Y '¢ "r /"é\$"Ù0"â " ("r " 8"ù@'R &"ù6"â . "y 'É." ( 'YK "™?)8'R 0")

## 49. What specific geometry defines an Inner Earthquake wave?

•i 'I0"ù 'ÙB'Y '¢ „-ææW" V thquake) 'I0•) 'YK 'YL', 8" 5"ù6"ù7"Ù 'ÉM'ù>'é?'I? '©0"ù-"é7"ù\$ 'Y0'I@ "T

### Meaning:

Transversal Wave (90 degrees).

•Y(" \*")Ù0"%M'R \$" 'r f" !"ù "Ù0" TMd

### Trick / KD Hack

Logic Link: Inner waves (secondary/shear waves) displace the earth's mantle vertically relative to their outward path, creating a 90-degree perpendicular intersection.

Logic Link: •i 'I0"ù 'I0•) "y ('iM"Y?'I@'ù /'Y\$" (" \$" 'yG•" "9%"Ù5" "r ."y 'ù2 'YK •TM('YG 'É>"TM0" \*'R "r 8"é\*y

## 50. The macroscopic visual of a Snake moving on the ground represents what kind of wave?

'É." ( '©0 '©2'IG "TMA•ò 8"é '¢ "â ."‰ "Ù0"18"Ù "1\*ù 'iC"iM'ò "ù8 '©M" "é0 'Y@ 'I0•) 'Y> '©M" \$"ù("ù"ù

### Meaning:

Transverse Wave.

•Y(" \*")Ù0"%M'R \$" 'r ...@ransverse Wave)–@

### Trick / KD Hack

Logic Link: A snake's ultimate goal is to move forward, but it accomplishes this by bending its body laterally (side-to-side) at a strict 90-degree angle to its forward path.

Logic Link: •ù "%>• \* 'Y> •Y 'I?'â 2'YM"yM'ò 'yG 'É""É("â 9",Â 2"y "ù( "Y9 •Y\*%G "i0" 0 'YK •Y\*%G •i "r "r \*'R 8

## 51. Flicking a Rope up and down creates what kind of wave?

" 8"Ù8" "2 '©0-%@'©G 'Ù 'Y("r 8"r "ù8 '©M" "é0 'Y@ 'I0•) •TM\$"Ù\*%M', 9"1\$" 9"f ð

### Meaning:

Transverse Wave.

•Y(" \*")Ù0"%M'R \$" 'r ...@ransverse Wave)–@

### Trick / KD Hack

Logic Link: The rope itself doesn't fly out of your hand across the room. The rope pieces simply move 90 degrees (up and down) to pass the energy forward down the line.

Logic Link: " 8"Ù8" " & •i\*'YG "TM>'R 8"r ' <'Y0 'Y." G 'YG •TM8 '©>" ("TM@•" "é\$" d " 8"Ù8" "r " ' <"r " M'É> 'Y